

Project Report – Best Weather App in the World Mate

Course: Python Flipped Course (VITyarthi Platform).

Component: Evaluated Course Project.

>Objectives

The aim of this project was to make a Python-based weather app that can fetch real-time weather data through an API and present it in a simple, user-friendly way directly in the command line.

Features Implemented

- Live weather data fetched from OpenWeatherMap .
- Command-line interface where user can put a city name.
- Error handling for invalid city names or API failures.
- Displays temperature and overall weather conditions.
- Secure API key management using a .env file.

Setup & Execution

Clone the repository:

```
git clone https://github.com/vyankteshgour-sudo/best-wather-app-in-the-world-mate
```

```
cd best-wather-app-in-the-world-mate
```

Install dependencies:

```
pip install -r requirements.txt
```

Create a .env file in the project root with:

```
WEATHER_API_KEY = fd17f1979692b6744a34e9ed52fa2d5
```

Run the app:

```
python main.py --city "London"
```

screenshots and demo

Screenshots are available in the screenshots/ folder and readme.

A screen recording demo is linked in the README.md.

Challenges Faced

- Handling API errors and invalid inputs and other errors nicely.
- Ensuring smooth execution via the command line.
- Managing environment variables securely.

Future Improvements

- Add support for fetching weather for multiple cities at once.
- Integrate forecast data (like next 5 days or week).
- Improve CLI output formatting with colours for better readability.
- Add unit tests to improve reliability and maintainability.